

ANSWER KEY — Integration Lesson 10 Test: Area Between Two Curves

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. Area between $y = 9$ and $y = 4$ from $x = 0$ to $x = 6$, checked with the rectangle formula.

Work: top = 9, bottom = 4, gap = 5. $\int_0^6 5 \, dx = 5x \big|_0^6 = 30 - 0 = 30$. Rectangle check: width 6 $\times$ height 5 = 30 ✓. **Answer: 30 square units.**

2. Area between $y = x + 7$ and $y = x + 3$ from $x = 0$ to $x = 5$.

Work: parallel lines; top is $y = x + 7$ (test $x = 0$: $7 > 3$). Gap = $(x + 7) - (x + 3) = 4$ — the x 's cancel. $\int_0^5 4 \, dx = 4x \big|_0^5 = 20$. **Answer: 20 square units.**

3. Area between $y = 10$ and $y = 2x$ from $x = 0$ to $x = 4$.

Work: top test at $x = 4$ (the biggest x here): 10 vs $2 \times 4 = 8$, so $y = 10$ stays on top on the whole interval. $\int_0^4 (10 - 2x) \, dx = (10x - x^2) \big|_0^4 = (40 - 16) - 0 = 24$. (Check: derivative of $10x - x^2$ is $10 - 2x$ ✓.) **Answer: 24 square units.**

4. Which is on top between 0 and 1: $y = x^2$ or $y = x^3$?

Work: test $x = 1/2$: x^2 gives $1/4 = 0.25$; x^3 gives $1/8 = 0.125$. Since $0.25 > 0.125$, the square is on top — multiplying a number less than 1 by itself again makes it even smaller, so higher powers sit lower on $(0, 1)$. **Answer: $y = x^2$ is on top ($0.25 > 0.125$ at $x = 1/2$).**

5. Area between $y = 9$ and $y = x^2$ from $x = 0$ to $x = 3$.

Work: top test at $x = 1$: 9 vs $1 \rightarrow$ line on top (they only meet at $x = 3$, since $3^2 = 9$). $\int_0^3 (9 - x^2) \, dx = (9x - x^3/3) \big|_0^3 = (27 - 9) - 0 = 18$. (Check: derivative of $9x - x^3/3$ is $9 - x^2$ ✓.) **Answer: 18 square units.**

6. Area of the region enclosed by $y = 3x$ and $y = x^2$.

Work: crossings: $3x = x^2 \rightarrow x^2 - 3x = 0 \rightarrow x(x - 3) = 0 \rightarrow x = 0$ and $x = 3$. Top test at $x = 1$: line $\rightarrow 3$, parabola $\rightarrow 1 \rightarrow$ line on top. $\int_0^3 (3x - x^2) \, dx = (3x^2/2 - x^3/3) \big|_0^3 = (27/2 - 9) - 0 = 27/2 - 18/2 = 9/2$. **Answer: $9/2$ square units (= 4.5).**

7. Runner A at 7 m/s, runner B at 4 m/s; A's lead after 20 seconds.

Work: the lead is the area between the two speed graphs. A's graph (height 7) is on top. Lead = $\int_0^{20} (7 - 4) \, dt = \int_0^{20} 3 \, dt = 3t \big|_0^{20} = 60$. Check: A ran 140 m, B ran 80 m, difference 60 ✓. **Answer: 60 meters.**

8. Faucet A at 8 L/min, faucet B at $2t$ L/min; when are the rates equal, and how far ahead is A then?

Work: rates equal: $8 = 2t \rightarrow t = 4$ minutes. A on top before then (test $t = 1$: 8 vs 2). Extra water = $\int_0^4 (8 - 2t) dt = (8t - t^2) \big|_0^4 = 32 - 16 = 16$. Check: A delivered 32 L, B delivered $t^2 \big|_0^4 = 16$ L, difference 16 ✓. **Answer: rates equal at $t = 4$ min; A is 16 liters ahead.**

9. Revenue rate 20 \$/day, cost rate $8 + 2t$ \$/day; total profit from $t = 0$ to $t = 6$.

Work: profit rate = $20 - (8 + 2t) = 12 - 2t$, which stays ≥ 0 on $[0, 6]$ (it reaches 0 exactly at $t = 6$). Profit = $\int_0^6 (12 - 2t) dt = (12t - t^2) \big|_0^6 = 72 - 36 = 36$. (Check: derivative of $12t - t^2$ is $12 - 2t$ ✓.) **Answer: \$36 profit.**

10. CHALLENGE: area enclosed by $y = x^2$ and $y = 2x + 3$.

Work: crossings: $x^2 = 2x + 3 \rightarrow x^2 - 2x - 3 = 0 \rightarrow (x - 3)(x + 1) = 0 \rightarrow x = -1$ and $x = 3$. Top test at $x = 0$: line $\rightarrow 3$, parabola $\rightarrow 0 \rightarrow$ line on top. $\int_{-1}^3 (2x + 3 - x^2) dx = (x^2 + 3x - x^3/3) \big|_{-1}^3$. At 3: $9 + 9 - 9 = 9$. At -1: $1 - 3 + 1/3 = -5/3$. Area = $9 - (-5/3) = 27/3 + 5/3 = 32/3$. (Check: derivative of $x^2 + 3x - x^3/3$ is $2x + 3 - x^2$ ✓.) **Answer: $32/3$ square units (≈ 10.67).**